

Project Report

Capstone 3: San Diego Development Opportunity Finder

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Business Problem + Project Goal

Finding a good development or investment opportunity requires you to look into a lot of different factors like the rental market, income, walkability, transportation, environmental factors, schools, etc. Doing that research tract by tract can get tedious pretty quickly, especially across an entire county.

I originally wanted to build something closer to a neighborhood scoring system inspired by tools like [Local Logic](#) which I rely on for any real estate research. Their location scores allow people to look at a variety of factors helpful with decision making. I didn't use any Local Logic data or copy their system, but my goal was to see what I could build independently using public data.

The project changed quite a bit as I worked through it. At first I wanted to build a tool that would create an overall “development opportunity score.” After getting a little deeper into the data, I found that relationships between features were really complex and it would require more resources to go really in depth. It was clear though, that there aren't really tracts that are “best” across everything, and different types of neighborhoods had different strengths.

I eventually reframed the project as a first-pass screening tool. Instead of trying to predict where developments might be successful, I use regression to estimate expected rent based on tract conditions (transit access, school scores, walkability, etc), then look for places where actual rent is lower than expected. I also use clustering to group similar tracts into broader neighborhood types (based on Claritas Segment Finder). It would be a first step for someone looking into development or investment opportunities, but would still require further research like market studies, leasing data, or financial feasibility. It helps narrow San Diego County census tracts into a smaller group that might be easier to work with.

Data Wrangling

This ended up being one of the biggest parts of the project. I combined demographic, housing, transportation, walkability, environmental, school, and safety information from several public sources. My overall strategy was to clean each source separately, convert everything to the census-tract level where possible, standardize tract IDs, create useful rates or scores, and then merge everything into one master dataset. I kept the original detailed files, but narrowed the final working dataset down to the features that were most useful for comparing tracts.

Census / ACS — Demographics + Housing

Source: <https://data.census.gov/table?q=DP03>

I used ACS data for income, poverty, unemployment, renter rates, vacancy, household size, age, commute behavior, and median gross rent. The raw Census tables had a lot of detailed columns, so I searched by topic, selected the fields I needed, renamed them, standardized tract IDs, and merged the smaller tables together.

Crime / Safety

Sources:

<https://data.sandiego.gov/datasets/police-nibrs/>

<https://data.sandiego.gov/datasets/gis-police-divisions>

The crime data used latitude/longitude instead of census tracts, so I spatially joined incidents to tract boundaries. I created violent and property crime rates and a safety score. I also found that very low-population tracts could distort crime rates, so I flagged those cases.

The larger issue was coverage. SDPD does not represent every law-enforcement agency in San Diego County, so I ultimately removed safety from the countywide modeling features. I kept `sdpd_jurisdiction_flag` so safety could still be used later as an additional screening layer.

Transit

Source: <https://data.sandiego.gov/datasets/gis-transit-stops>

I spatially matched transit stops to census tracts and created features such as stop count and transit-stop density. Density was more useful than raw stop count because it made tracts of different sizes easier to compare.

Walkability

Source: <https://www.epa.gov/smartgrowth/smart-location-mapping#walkability>

The EPA walkability data had a major geography issue because its tract boundaries did not fully match the newer Census geography. About 199 tracts were missing values. Instead of dropping almost 25% of the data, I rebuilt those values using smaller Census block-group scores inside the tracts, weighted by household count. I kept a flag showing which walkability values were estimated.

Climate / Environmental Risk

Source: <https://resilience.climate.gov/datasets/FEMA%3A%3Aational-risk-index-census-tracts/about>

I used FEMA National Risk Index data to bring in climate-loss, heat, flood, wildfire, and social-vulnerability measures. Most of this data was already tract based, so the main work was selecting useful fields and renaming them into clearer risk scores.

Schools

Sources:

<https://data.ca.gov/dataset/california-public-schools-2024-25>

<https://data.ca.gov/dataset/california-school-district-areas-2024-25>

<https://www.cde.ca.gov/Ta/ac/cm/acaddatafiles.asp>

I counted school locations by tract, calculated school density, and added district-level academic information. The hardest part was that school-district and census-tract boundaries do not line up exactly. I calculated the amount of each tract that overlapped each district and assigned the district with the largest overlap. This gives useful school context, but it does not tell me the exact elementary, middle, or high school assigned to every tract.

Final Merge + Missing Data

After merging everything, the full dataset had more than 200 columns. I kept that version for reference, then created a smaller dataset focused on the variables that were useful for EDA and modeling. Which brought about different problems that I didn't anticipate earlier. Different source years created geography mismatches, some features were missing, and not every dataset had the same geographic coverage.

The two biggest missing-data issues were walkability and rent. I rebuilt the 199 missing walkability values using household-weighted block-group data. **median_gross_rent** had 79 missing values, so I inspected renter rates, income, and the affected tracts before creating **estimated_median_gross_rent**. I estimated missing rents using related tract groups and kept a flag showing which original rents were missing.

During preprocessing, I ran into the issue of missing household income values that I left null during data wrangling. I ended up imputing a median value from related base tracks (ie: 110.1, 110.2, 110.3, etc) rather than dropping those tracts since I wanted to retain as much data as possible for the other features. I figured this approach would be more accurate than using a countywide average.

The final dataset had 727 census tracts with a complete modeling target.

Exploratory Data Analysis (EDA)

(figures, visualizations, feature importance and relationships)

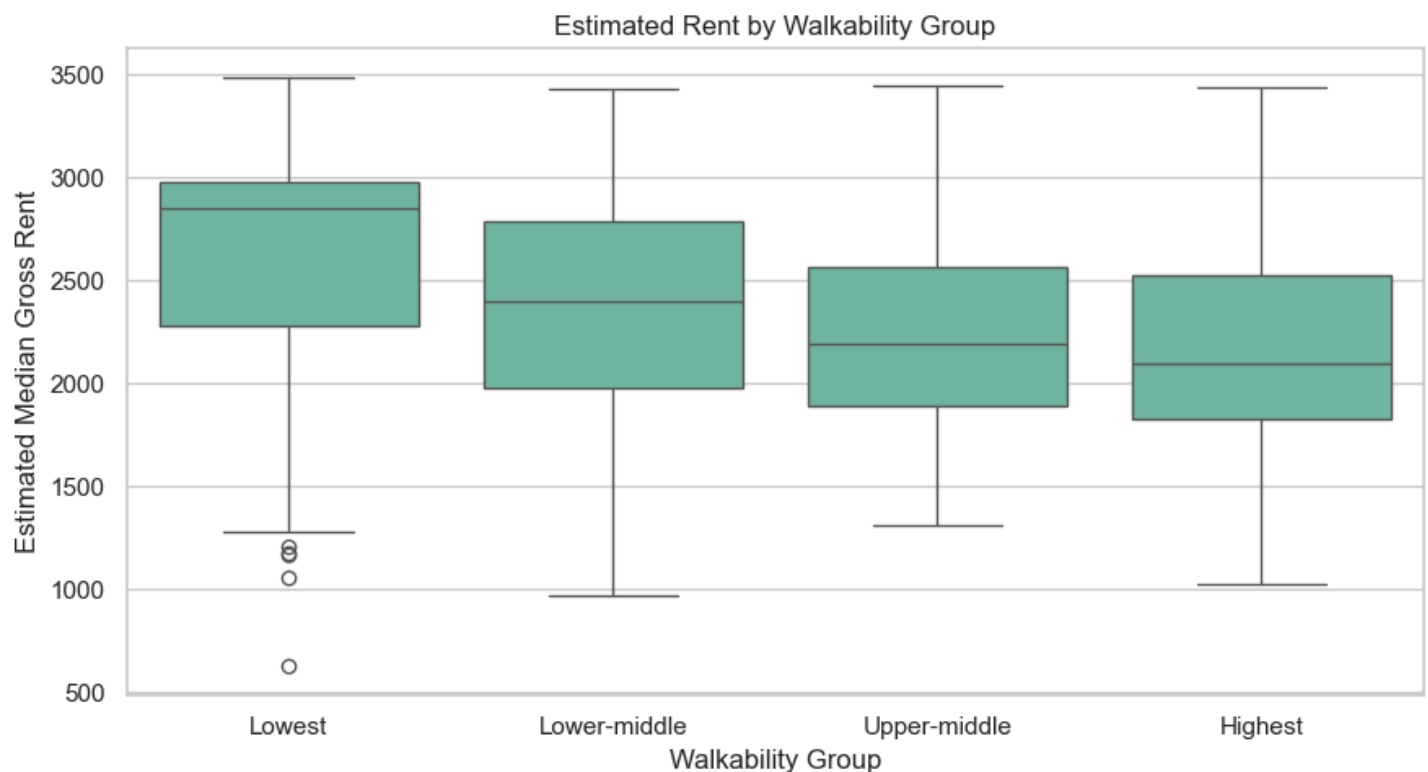
EDA helped me understand which features moved together, where there were unexpected patterns, and which parts of the original project idea needed to change. I used distributions, scatterplots, grouped boxplots, correlation heatmaps, maps, and tract comparisons to explore relationships before modeling.

Some findings were expected, like higher household income generally lining up with higher rent and poverty moving in the opposite direction. Other patterns were more interesting. Lower-walkability tracts tended to have much higher wildfire risk, while flood risk was spread more evenly across walkability groups. Higher school academic scores generally lined up with higher rents, but there were also some unexpected poverty and income outliers.

Family-heavy tracts developed their own pattern: larger households, lower vacancy, less transit, and somewhat lower walkability. That became useful later when I created neighborhood clusters.

EDA also showed me that no tract was clearly “best” across every feature. That was one of the reasons I moved away from creating one overall opportunity score and toward using regression and clustering together.

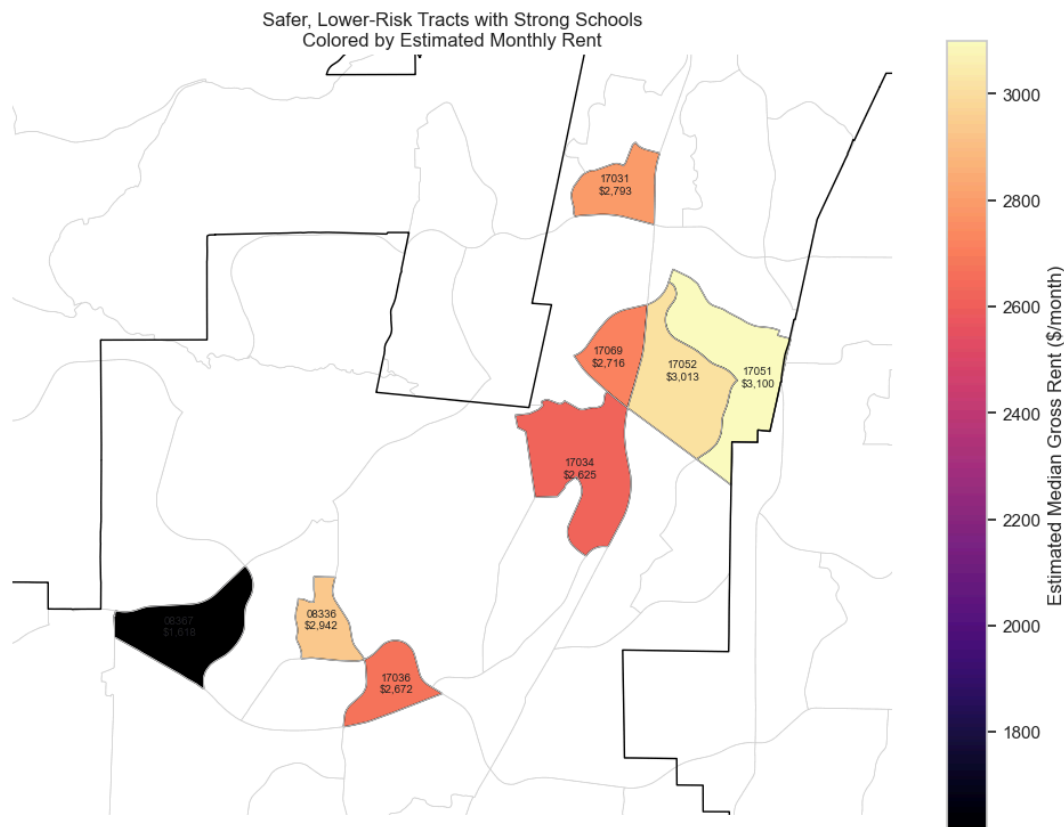
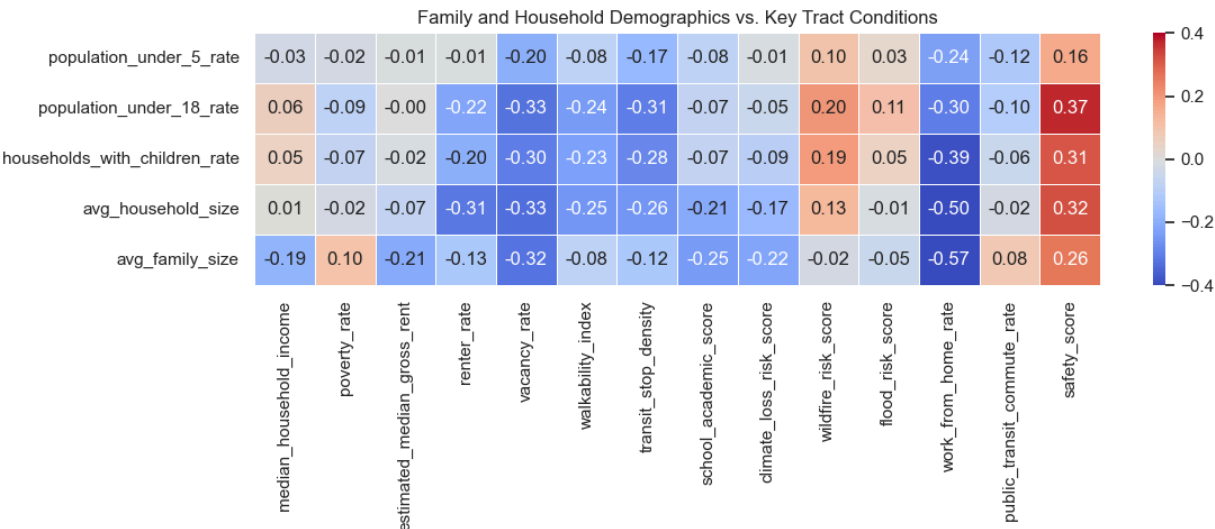
Visuals:



Key Points:

- The lowest-walkability group has the highest median rent.
- Median rent drops as walkability increases.
- The highest-walkability group has the lowest median rent.
- There is still a lot of overlap, so walkability is not the only factor affecting rent.

Feature Correlation Heatmap



I used percentile cutoffs to narrow the SDPD-covered tracts to the top 25% for safety and bottom 25% for climate-loss risk. I then mapped the remaining tracts with stronger school scores and colored them by estimated rent. This shows how the dataset could be used to quickly turn a large countywide dataset into a smaller group of areas worth looking into more closely.

Dummy Features

I didn't create dummy variables for the final modeling dataset since all 20 final model features were numeric. If I included something categorical in future versions, like district name, neighborhood type, or land-use category, I would encode them before using them in regression.

Standardization + Preprocessing

Before modeling, I narrowed the data down to 20 numeric features covering transportation, walkability, climate, schools, demographics, household characteristics, and housing conditions.

Two features needed log transformations because their distributions were skewed. :

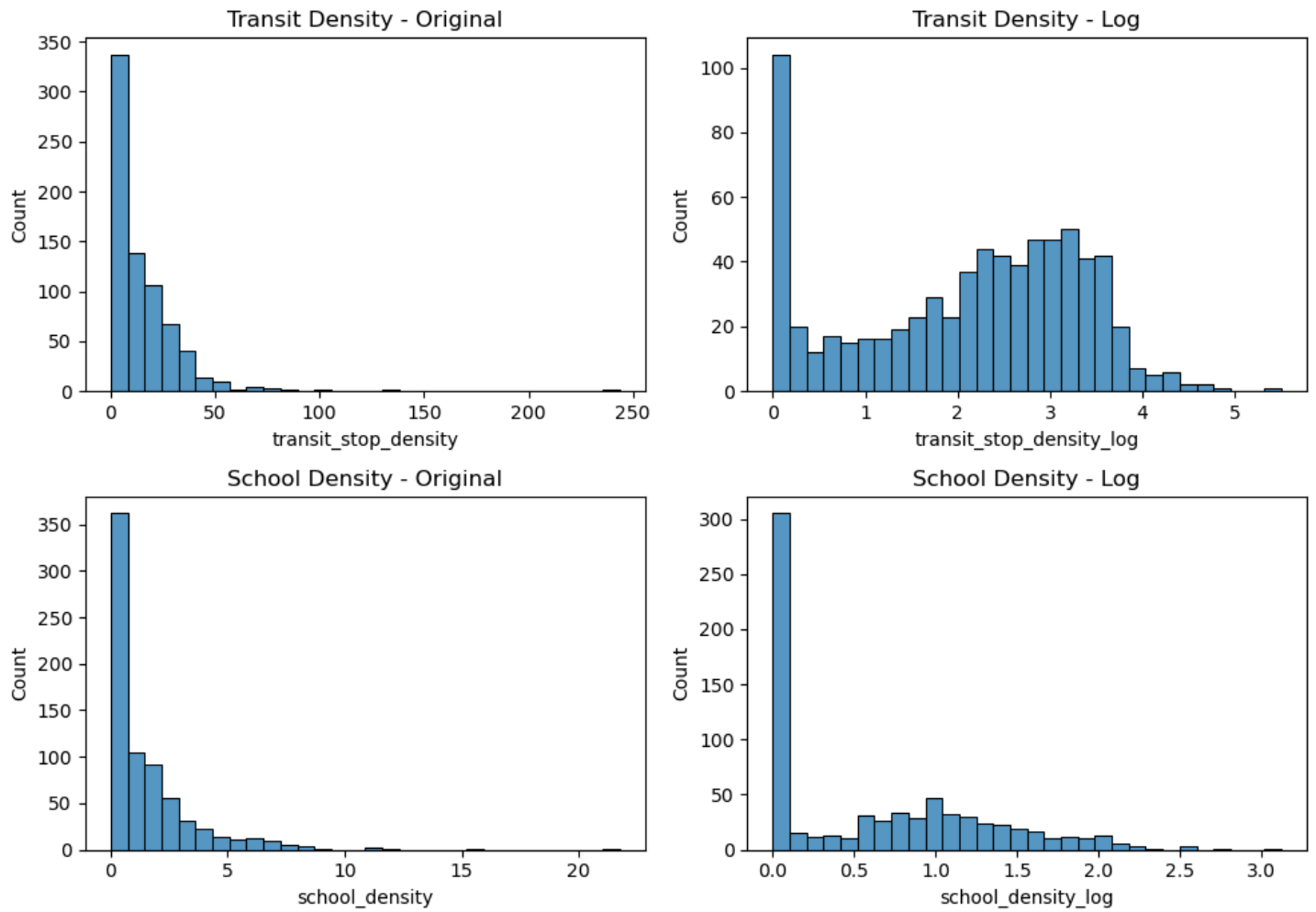
- **transit_stop_density** → **transit_stop_density_log**
- **school_density** → **school_density_log**

I used **log1p()** for both. Transit-density skew dropped from **4.39 to -0.34**, and school-density skew dropped from **3.05 to 0.71**.

After that, I standardized the modeling features with **StandardScaler** because features were measured on different scales. For example, household income is measured in dollars, but school scores and risk scores use much smaller ranges.

The scaler was fit **only on the training data** and then applied to the test data. I also saved the fitted scaler with **joblib** so I could use that transformation later.

Visual:



I filled the remaining eight missing household-income values using related tract information rather than dropping those tracts.

Train Test Data Split

I used an 80/20 train-test split.

The final split was:

- Training: 581 tracts × 20 features
- Test: 146 tracts × 20 features
- Total: 727 tracts

The train/test split was created before fitting the scaler or any regression models to help prevent leakage into model training.

Modeling

I used two supervised regression models to predict **estimated_median_gross_rent** and one unsupervised model to group similar tracts.

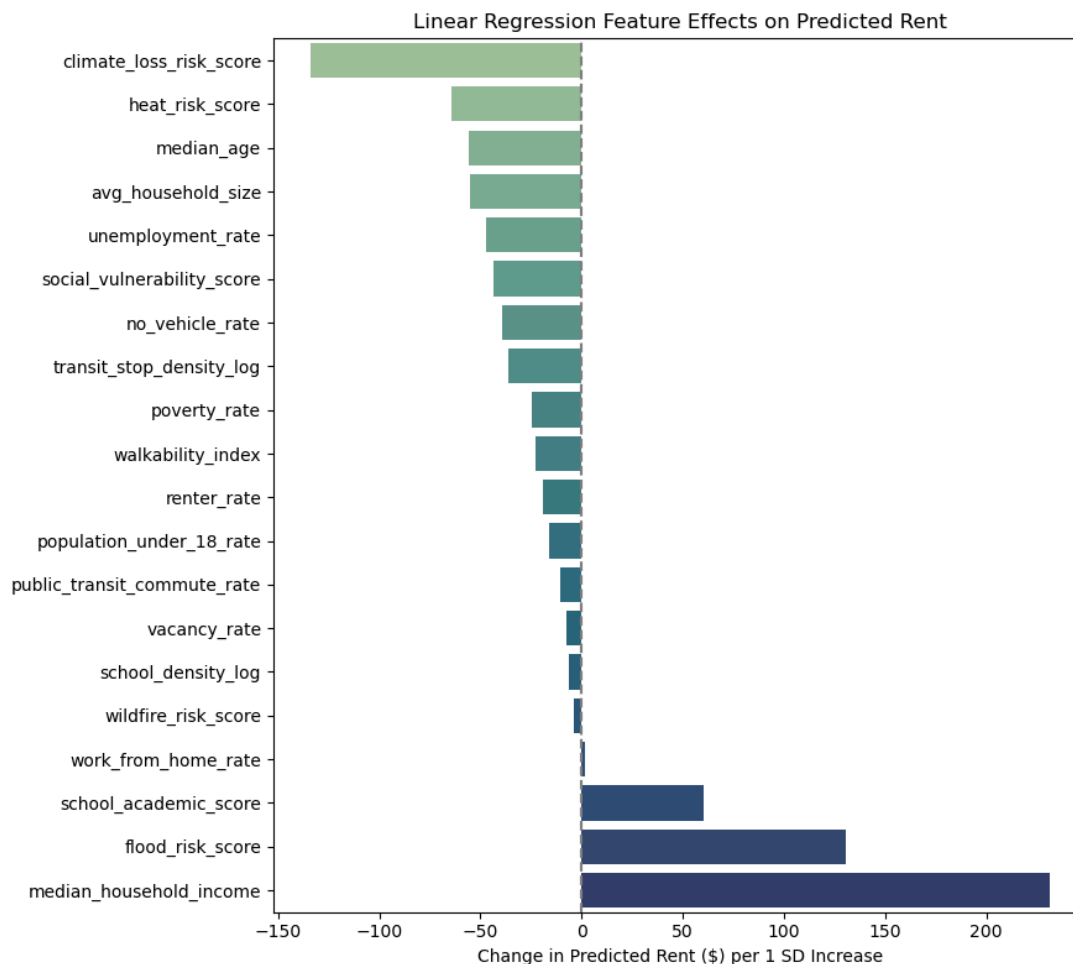
- **Linear Regression:** My baseline model for predicting expected rent. It made it easy to compare which features had the strongest positive or negative relationship with rent.
- **Random Forest Regressor:** I tested this to see if more complicated nonlinear relationships improved rent prediction. The original version overfit the training data, so I tuned it using **GridSearchCV** with 5-fold cross-validation on the training set only. The best version used **500 trees, max depth 12, and minimum 10 samples per leaf**.
- **K-Means Clustering:** Used separately from regression to group all 727 tracts into similar neighborhood types. I tested K=2 through K=8 using inertia and silhouette score and selected **K=3** as the best balance between separation and useful interpretation.

Modeling Results + Comparison

	model	train_r2	test_r2	test_rmse	test_mae
0	Linear Regression	0.504	0.566	364.386	277.743
1	Tuned Random Forest	0.696	0.535	377.405	281.137

Linear Regression performed slightly better on every test metric, so I kept it as the final regression model.

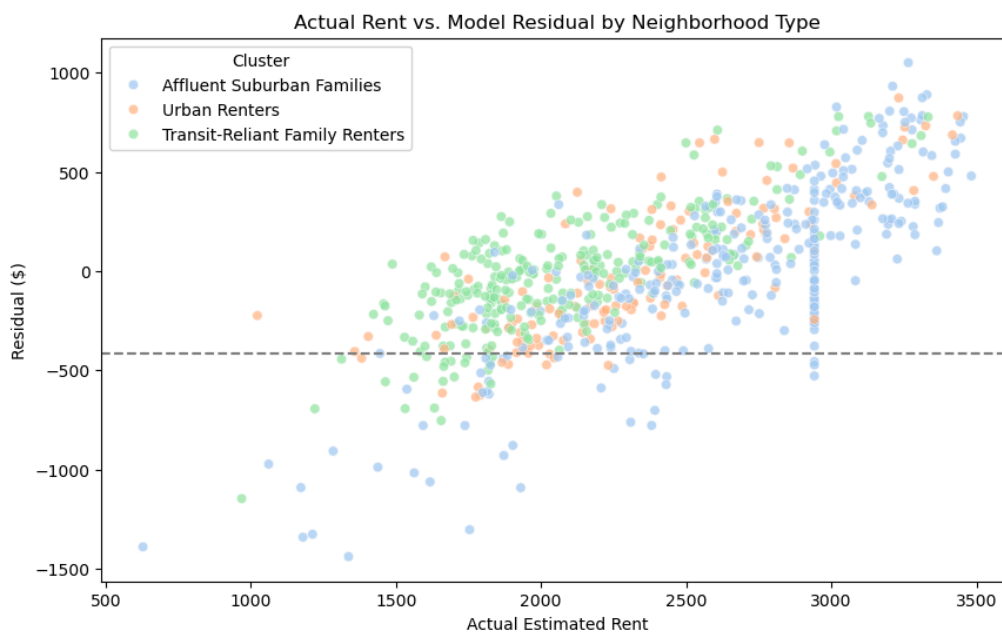
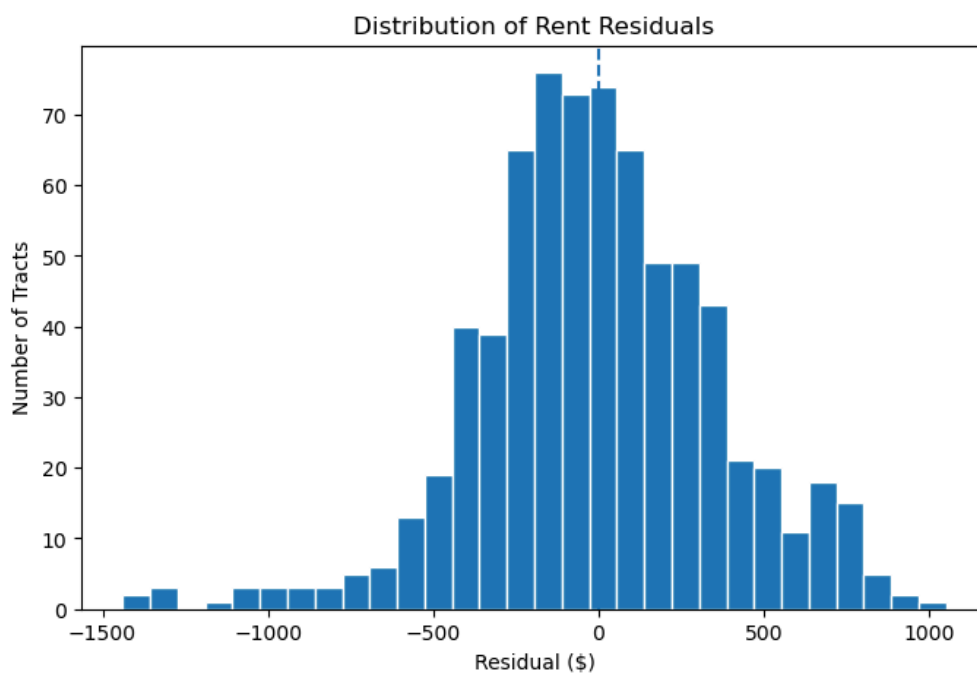
Feature Importance / Feature Relationships in the Final Model



Bars on the left side of zero are associated with lower predicted rent, while bars on the right are associated with higher predicted rent. The longer the bar, the stronger an effect it has. The same is true for the right side, but the higher numbers mean rental prices go up. I think this is an interesting interpretation of the data, but I'd be curious to change the target variable and median income to see what other underlying relationships exists.

Residual Screening

After testing the final Linear Regression model, I used it to estimate rent for all 727 tracts and compared those predictions with actual rent. I used the bottom 10% of residuals as a screening cutoff, which was about -\$415. This flagged 73 tracts where actual rent was much lower than expected and worth looking into more closely.



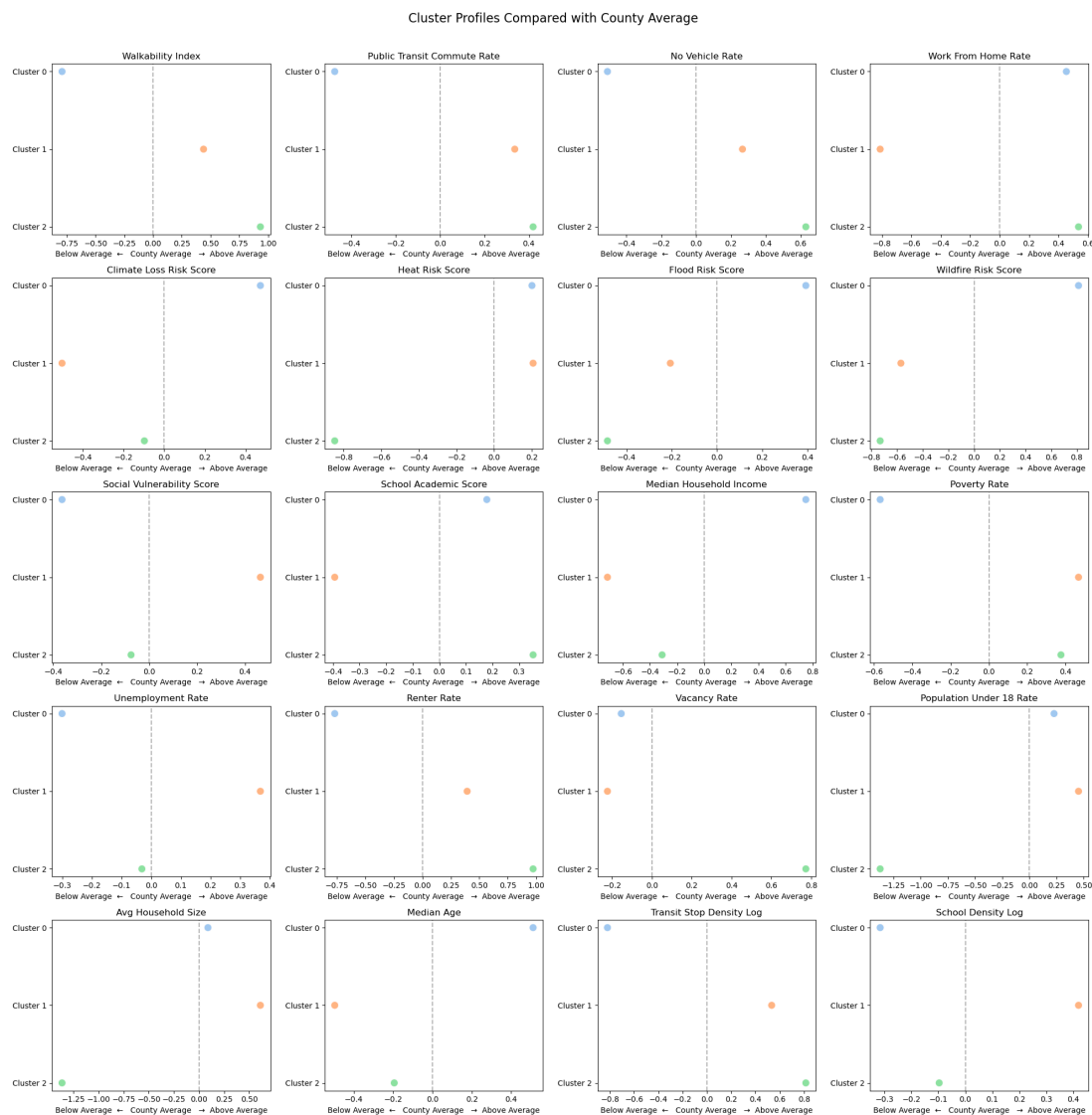
Lower-than-expected rents showed up across all three neighborhood types, so the residual screening wasn't limited to one kind of tract. Affluent Suburban Family tracts had the highest share of flagged tracts, around 13%, compared with about 8% in the other two groups.

Unsupervised Learning (Clustering) Results:

The three clusters gave me broad neighborhood types, but the silhouette scores were low (**K=2** $\approx$ **0.183**, **K=3** $\approx$ **0.181**), so these are general profiles with overlap rather than clean-cut neighborhood categories :

- **Affluent Suburban Families (316 tracts):** higher income, larger households, lower transit/walkability, and higher wildfire risks.
- **Transit-Reliant Family Renters (270 tracts):** lower income, more renters and children, higher social vulnerability, and higher transit dependence.
- **Urban Renters (141 tracts):** highest walkability, renter rates, transit density, and no-vehicle rates, with smaller households.

These subplots shows where each cluster landed in terms of countywide averages for each feature



Model Analysis + Business Use

The model's R2 was around 57% which means it would not be useful in predicting pricing, but it can be used as a screening tool not for decision making. I would use it to identify unusual tracts, especially places where actual rent is much lower than expected based on the features I included.

From a management perspective, I'd use the flagged tracts to reduce the amount of manual research needed. Instead of reviewing all 727 tracts, a team could start with the 73 strongest negative residuals, then apply filters based on the type of project they're considering.

The next step would still require more detailed research like zoning, parcel availability, land cost, construction feasibility, rent comps, permitting, and site visits.

I'd also recommend improving the model before using it for higher-stakes decisions by adding features like unit size, housing age and condition, zoning, coastal proximity, and market trends.

The model is valuable because it helps narrow down areas by comparing key features and highlighting places that deserve closer review, instead of manually doing all of the research.

Further Research:

1. **Explore feature relationships in depth:** Remove rent and income features to see how different scores interact with each other. For example, do school scores increase or decrease with walkability or transit access?
2. **Use more demographic features:** For this version, I limited the amount of features from the Census dataset, but in another version I'd look at demographic data more closely (age, education levels, ethnicity, etc) to see how they are dispersed among tracts. I'd use a clustering model to see what other features move together with demographic features.
3. **Education:** The original version did not have schools assigned to each tract based on district boundaries. I would spend more time to find which elementary, middle, and high schools are assigned to each tract by merging the geo data and manually assigning the schools to the tracts if necessary. Then I'd be able to get a more accurate score of education scores per tract and see if there are specific qualities/features of a tract that contribute to higher education scores. The goal would be to find out what neighborhood features contribute to high scores and then find opportunities to improve them.
4. **Development Analysis:** I didn't pull any permit data or new developments for this project, but I am interested in seeing commonalities between new developments around San Diego. It would take a

lot more research and analysis to see the success of new developments (leasing rates, financial data, etc), so I think I'd have to limit my scope to just see similar traits among new products.

5. **SANDAG Data:** SANDAG has a large data catalog ranging from transportation, parcels/geographic boundaries, crime data, census data, and community data. I think for a more tailored and accurate modeling project, I would directly pull from these databases and look into other GIS based problems.

Client Recommendations:

1. **Site Screening:** Use as a screening tool to quickly find tracts that might suit a development product (residential, educational, transit, etc)
2. **Opportunity Identification:** Find tracts that look undervalued based on predicted rent compared with actual rent, which could point to areas worth looking into.
3. **Neighborhood Comparison:** Compare similar tracts with similar scores for walkability, transit, schools, demographics, safety, and environmental to see which could benefit (increase scores) for different features.